Higher-MI Predictor: Method and Results

A step-by-step explanation of the benchmark adaptation

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Purpose and principal result

This document describes the implementation and evaluation of Higher-MI as an early-warning predictor of *grokking* in a modular-addition task. Grokking is delayed generalisation: a model first fits its training examples, and only after many additional training epochs does it generalise to unseen examples.

The tested predictor uses three summaries of the model's output logits and estimates their population O-information across the test set at each saved checkpoint. The event used for scoring is the global maximum of this information curve.

Result. Under the benchmark's fixed scoring protocol, Higher-MI fails as an early-warning predictor. In all five runs, the global maximum occurs after the grok epoch. The measured scores are:

- early-warning fraction: $0/5 = 0$;
- Spearman rank correlation between grok epochs and peak epochs: $\rho = 0.7$ (required: at least 0.9);
- mean relative lead: -0.4254 (the peak occurs late, on average).

The curve also contains repeated spikes. Its first spike above 0.2 occurs at epoch 1,900 in every run, although the runs generalise at substantially different epochs. This is evidence about this implementation and benchmark only. It does not establish that the underlying information-theoretic idea is generally ineffective, and it does not contradict the source paper's result in its original setting.

The sections proceed in order: task definition; logits; the three logit summaries; the information measure; its numerical estimation; the observed results; and the assumptions and implementation details.

Step A. Task definition and grokking

Modular addition

The model receives two integers a and b and predicts

$$c = (a + b) \bmod 113. \tag{1}$$

Both inputs belong to $\{0, 1, \ldots, 112\}$, and the answer also belongs to this set. For example,

$$(23 + 45) \bmod 113 = 68,$$
$$(100 + 20) \bmod 113 = 120 - 113 = 7.$$

There are $113^2 = 12{,}769$ possible input pairs and 113 possible output classes.

Training set and test set

The input pairs are randomly split into two disjoint sets.

Set	Number of pairs	Use
Training set	3,830 (30%)	Used to update the model's weights.
Test set	8,939 (70%)	Not used for training; used to measure generalisation.

Training accuracy measures performance on the training set. Test accuracy measures performance on input pairs not shown during training. A high training accuracy alone therefore does not establish that the modular-addition rule has been learned. In seed 0, the pair $(23, 45)$ is a test example.

Epochs, random seeds, and checkpoints

Training runs for 40,000 epochs. In each epoch, the model processes all 3,830 training pairs once and performs a weight update. A random seed fixes the initial weights and the train/test split. Five seeds, numbered 0 through 4, are used.

The model weights are saved every 100 epochs. A saved state is called a *checkpoint*. Each seed has 401 checkpoints: epochs $0, 100, \ldots, 39{,}900$, plus the final checkpoint at epoch 40,000. All analyses in this document use these saved states; the model was not retrained for the analysis.

Definition of the grok epoch

The training accuracy reaches at least 0.99 near epoch 205 in all five runs. Test accuracy remains low for a much longer period and then rises. We define the grok epoch G as the first epoch at which test accuracy exceeds 0.9:

$$G = \min\{t : \mathrm{Acc}_{\mathrm{test}}(t) > 0.9\}. \tag{2}$$

The observed grok epochs are listed in Table 1.

Table 1: Grok epoch for each seed.

Seed	0	1	2	3	4
G	12,987	6,921	9,510	11,019	23,747

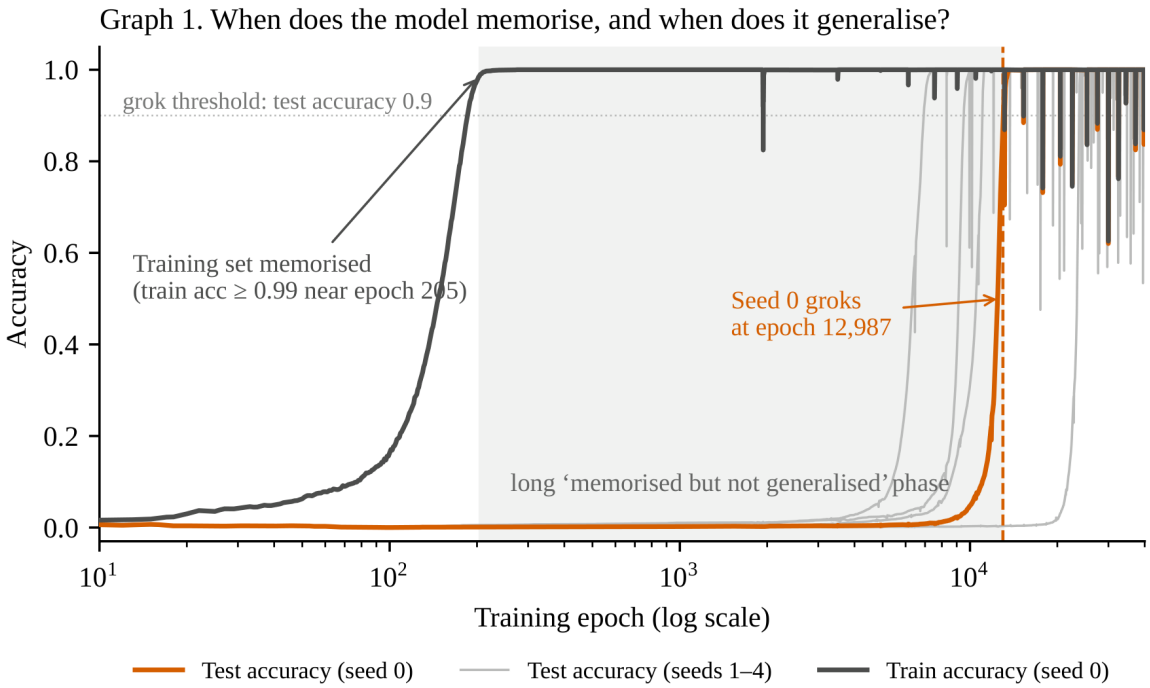


Figure 1: Original accuracy plot. Training accuracy rises early, while test accuracy rises much later. The plot and its annotations are reproduced from the source document.

The grok epochs differ considerably across seeds. A useful early-warning signal should occur before G in each run and should occur later in runs with later grok epochs. A signal that appears only at or after G is not an early warning.

Step B. Model outputs: logits

For an input pair such as $(23, 45)$, the model produces a real-valued score for each candidate answer $j \in \{0, \ldots, 112\}$. The score for answer j is the *logit* $\ell[j]$. Thus, one input produces a vector of 113 logits.

The following properties will be used:

1. A logit is a raw, unnormalised score. The logits are not required to sum to one.
2. A larger logit gives that answer a higher rank than an answer with a smaller logit.
3. A logit is not a probability. It may be negative or greater than one. Applying softmax would produce probabilities, but softmax is not used in this Higher-MI calculation.
4. The predicted answer is the index of the largest logit.
5. The correct answer is known from the modular-addition rule, so the prediction can be compared with the correct class.

In the trained network, the logits are typically around 30 to 150. The information estimators used below are invariant to a change of scale, so this range is not itself part of the predictor definition.

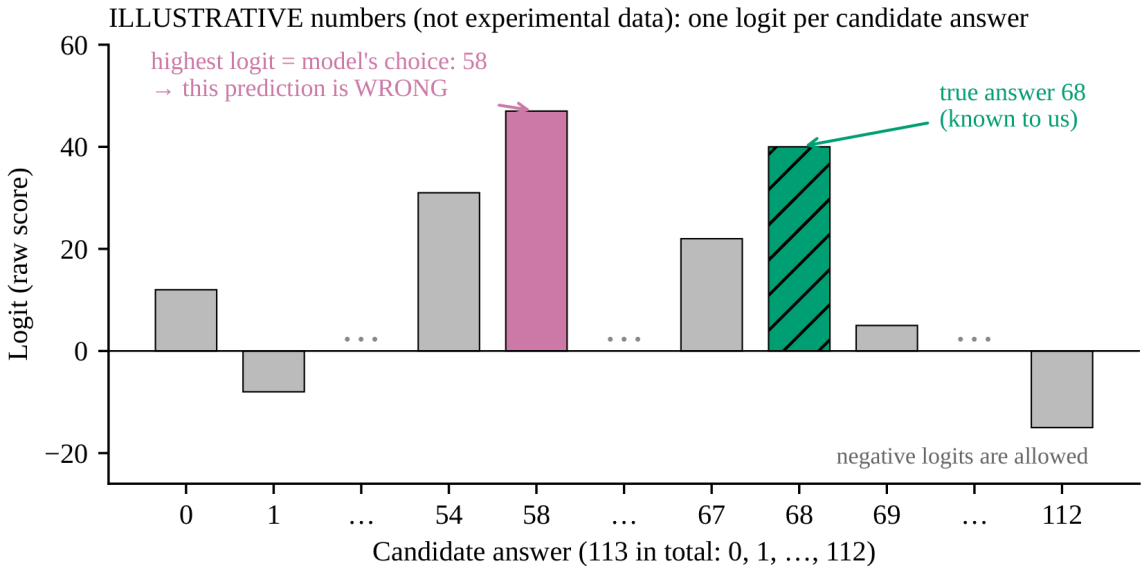


Figure 2: Original illustrative logit vector. The values in this diagram are synthetic and are not experimental measurements.

Step C. Reducing 113 logits to three variables

Higher-MI, as implemented here, does not use the full 113-dimensional logit vector. It reduces the vector to three real-valued variables for each input example. Let c denote the correct class and define the runner-up class as

$$r = \arg\max_{j \neq c} \ell[j]. \tag{3}$$

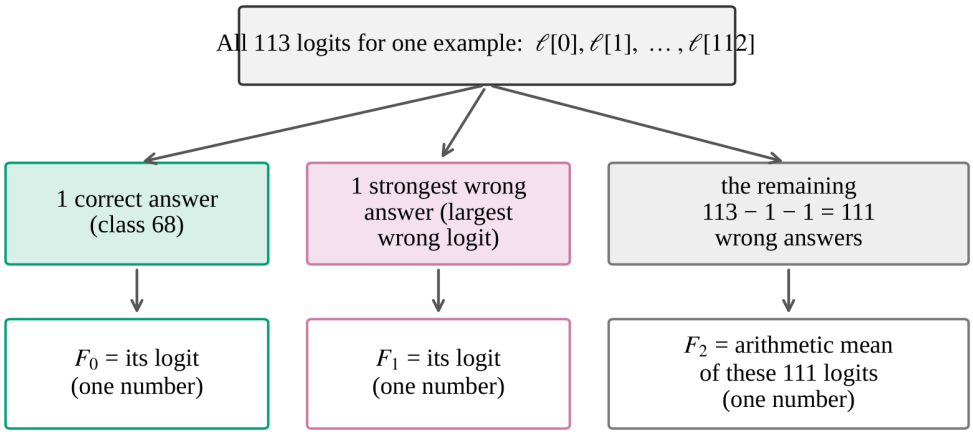
The three summaries are as follows.

$$F_0 = \ell[c], \tag{4}$$

$$F_1 = \max_{j \neq c} \ell[j] = \ell[r], \tag{5}$$

$$F_2 = \frac{1}{111} \sum_{j \notin \{c, r\}} \ell[j]. \tag{6}$$

Here, F_0 is the correct-answer logit, F_1 is the largest logit among the 112 incorrect answers, and F_2 is the mean logit of the remaining 111 incorrect answers after excluding the correct class and the runner-up. F_2 is an arithmetic mean; it is not a probability or a calibrated confidence score.



How the 113 logits of one example become three numbers.

Figure 3: Original diagram showing how one vector of 113 logits is reduced to F_0, F_1, and F_2.

The model is correct exactly when the correct-answer logit is greater than every incorrect-answer logit. Equivalently,

$$\text{prediction is correct} \iff F_0 > F_1. \tag{7}$$

The difference

$$m = F_0 - F_1 \tag{8}$$

is the classification margin. A positive margin indicates a correct prediction; a negative margin indicates an incorrect prediction, apart from the tie case.

Example at three checkpoints

For the test input $(23, 45)$, the correct answer is 68. At epoch 5,000, class 58 has the largest logit and the margin is about -36.4; the prediction is incorrect and F_2 is above F_0. At epoch 12,900, class 68 has the largest logit, with margin $+5.4$ and class 54 as the runner-up. At epoch 20,000, the margin is about $+15.8$.

The real example at three checkpoints

Graph 4. One unseen example, (23, 45) → 68, seed 0: how the three scores change

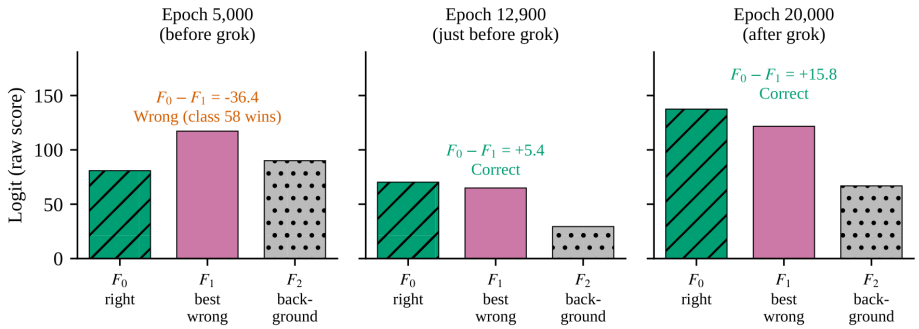


Figure 4: Original three-checkpoint example for the input $(23, 45)$ in seed 0. Bar heights and annotations are retained from the source figure.

This example illustrates the definitions of the three summaries. It does not by itself establish how the predictor behaves across the complete test set.

Step D. Reason for the three-variable reduction

The Higher-MI method is based on work by Pomarico et al. (2025, arXiv:2507.23346). Their study used a matrix product state classifier, a tensor-network model, on three-class tasks: a reduced Fashion-MNIST task and hyperspectral land-cover data. The model in that setting provides three class scores, and the information measure is defined for three variables.

The task in this benchmark has 113 output classes. The reduction in Equations (4)–(6) is therefore a project-specific adaptation: it retains the score of the correct class, the largest incorrect score, and the mean of the other incorrect scores. The source paper does not prescribe this reduction.

This reduction discards information. It does not retain the identities of all incorrect classes, the distribution of the 111 remaining logits, or the number of incorrect classes with scores close to F_1. Another reduction could produce another information curve. This is one of the experiment's assumptions.

The network's output layer has 114 entries: 113 numeric answer tokens and one additional entry for the token "=". Since "=" is not a valid answer class, it is not included in the calculations of F_0, F_1, and F_2. Including it could alter the runner-up or the mean.

Step E. Calculation over the complete test set

The three summaries are computed separately for every test example at a fixed checkpoint t. This gives 8,939 triplets

$$(F_0^{(i)}(t), F_1^{(i)}(t), F_2^{(i)}(t)), \qquad i = 1, \ldots, 8{,}939.$$

The KSG-based information calculation described in Step G is then applied to these triplets and produces one value, $\Omega_{\text{pop}}(t)$, for that checkpoint. Repeating the calculation at all 401 checkpoints produces one curve for each seed.

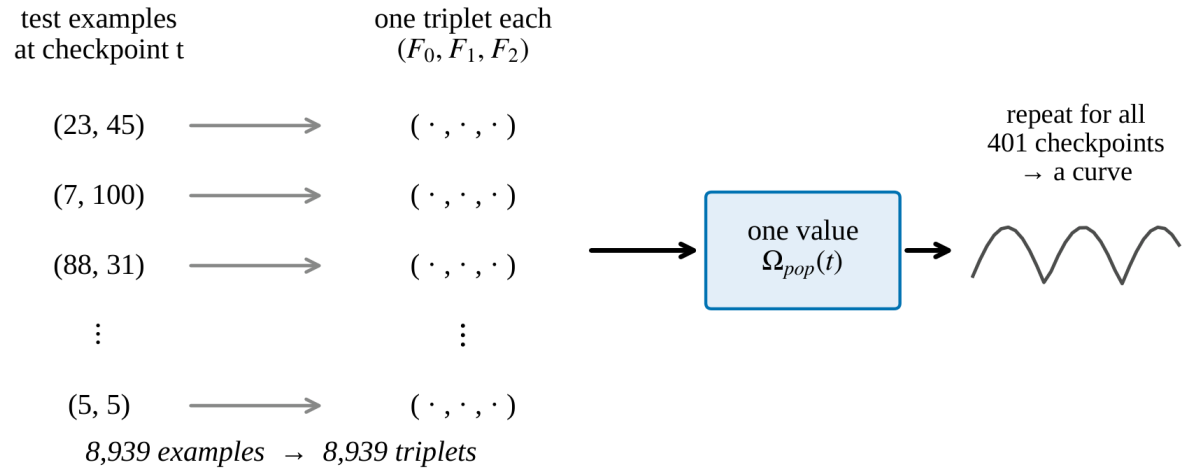


Figure 5: Original workflow diagram: every test example gives one triplet, the triplets give one value per checkpoint, and the checkpoint values form a curve.

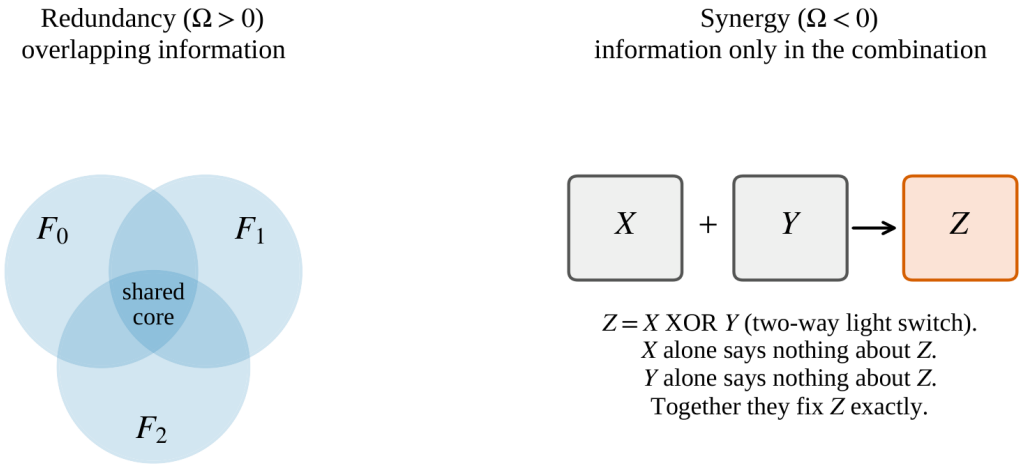
Across the test set, the triplets vary because different examples are easier or harder at a given checkpoint. The fraction of test triplets satisfying $F_0 > F_1$ is exactly the test accuracy. The test set is used because grokking is defined here through performance on unseen inputs; the training set reaches high accuracy much earlier. Using test examples is necessary for the question being tested, but does not guarantee that a signal will predict the grok epoch.

Step F. Mutual information and O-information

Redundancy and synergy

Three variables may contain overlapping information or information that is available only when variables are considered jointly.

- **Redundancy:** variables contain overlapping information. Knowing one variable may make the others partly predictable.
- **Synergy:** a relationship becomes informative only when variables are considered together. For example, if $Z = X \oplus Y$ is the exclusive-OR operation on two binary variables, then neither X nor Y alone determines Z, but the pair (X, Y) does.



Schematic. Left: overlapping information. Right: information present only in the combination.

Figure 6: Original redundancy and synergy schematic.

Mutual information

Mutual information $I(X;Y)$ measures the statistical dependence between variables X and Y. It can be interpreted as the reduction in uncertainty about Y obtained by observing X. It is zero for independent variables and is measured in nats when natural logarithms are used. It captures nonlinear dependence as well as linear dependence.

Conditional mutual information, $I(X;Y \mid Z)$, measures the dependence between X and Y after conditioning on Z; equivalently, it measures how much information about Y is supplied by X when Z is already known.

O-information

For the three logit summaries, O-information is defined as

$$\boxed{\Omega(F_0, F_1, F_2) = I(F_0; F_1) - I(F_0; F_1 \mid F_2).} \tag{9}$$

The first term measures the dependence between the correct-answer logit and the largest incorrect-answer logit. The second measures the dependence remaining after conditioning on the background mean F_2.

The sign convention from the source paper is retained:

- $\Omega > 0$: redundancy dominates;
- $\Omega < 0$: synergy dominates.

The sign does not directly classify the model as having or not having grokked. O-information describes a statistical relationship among the three summaries across examples. It is not itself a test-accuracy measure. For example, at epoch 20,000, F_0 and F_1 have a correlation of 0.998 across examples, whereas Ω is about 0.36 nats. High correlation does not require a large value of O-information.

Population O-information and the hypothesis

At checkpoint t, the primary measure $\Omega_{\text{pop}}(t)$ is calculated across the 8,939 test-example triplets. It describes the relationships among the summaries across examples at one fixed checkpoint. It does not describe how the summaries vary over training time.

The source paper reports strongly negative O-information at initialisation, a change to positive values as learning proceeds, and a redundancy peak near grokking in its setting. This benchmark tests the stronger early-warning hypothesis:

> The global maximum of $\Omega_{\text{pop}}(t)$ occurs before G, and the peak timing follows the variation in G across seeds.

Step G. Numerical estimation of O-information

Why an estimator is required

Mutual information is defined in terms of probability distributions, but the distributions of F_0, F_1, and F_2 are not known. Only 8,939 sample triplets are available at each checkpoint. A histogram approach would divide each axis into bins and count observations, but the estimated information can change with the chosen bin width.

The Kraskov–Stögbauer–Grassberger (KSG) estimator instead uses distances between sample points. For each point, it finds the distance to its fourth-nearest neighbour in the joint space. It then counts the observations within the corresponding width in lower-dimensional spaces. The comparison between joint-space and marginal-space counts is used to estimate mutual information.

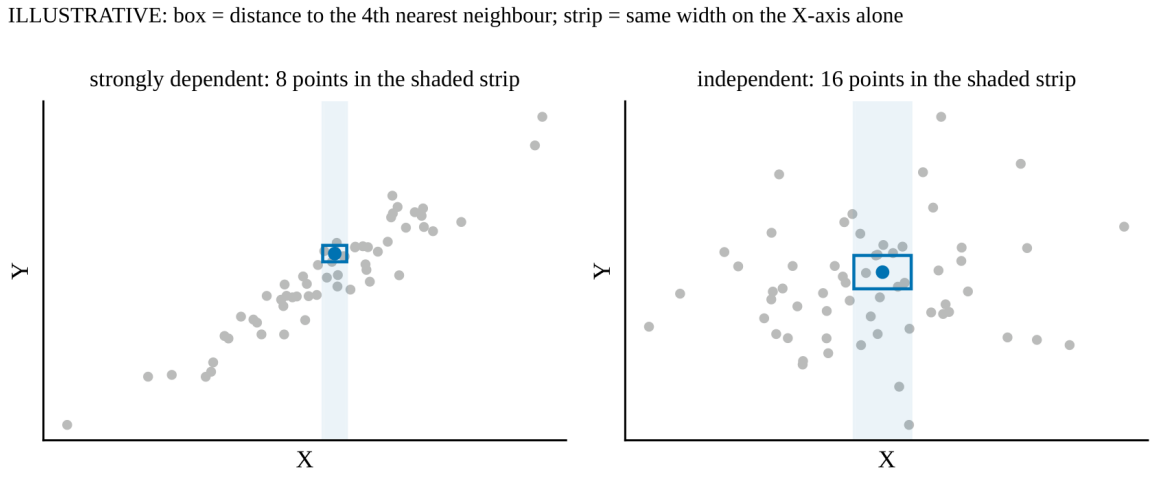


Figure 7: Original illustrative KSG diagram. The point configurations are synthetic and are not experimental data.

The experiment uses $k = 4$. Although the theoretical O-information expression is symmetric in the three variables, the finite-sample estimator need not be exactly symmetric. Therefore, the calculation is performed for all six orderings of (F_0, F_1, F_2) and the estimates are averaged. The formulas and counting convention are given in Appendix A.

Estimator checks

The implementation was checked against known-answer cases: independent variables should give mutual information near zero; near-duplicate variables should produce positive O-information; an XOR-like construction should produce negative O-information; and a synthetic logit vector with manually chosen values should yield the expected F_0, F_1, F_2 while ignoring the "=" output.

These checks support the correctness of the measurement implementation. They do not establish that the measured signal predicts grokking; that is evaluated separately by the timing protocol.

Step H. Results and timing evaluation

Event definition

Each seed has a population O-information value at 401 checkpoints. The predictor event is defined as the checkpoint with the global maximum value:

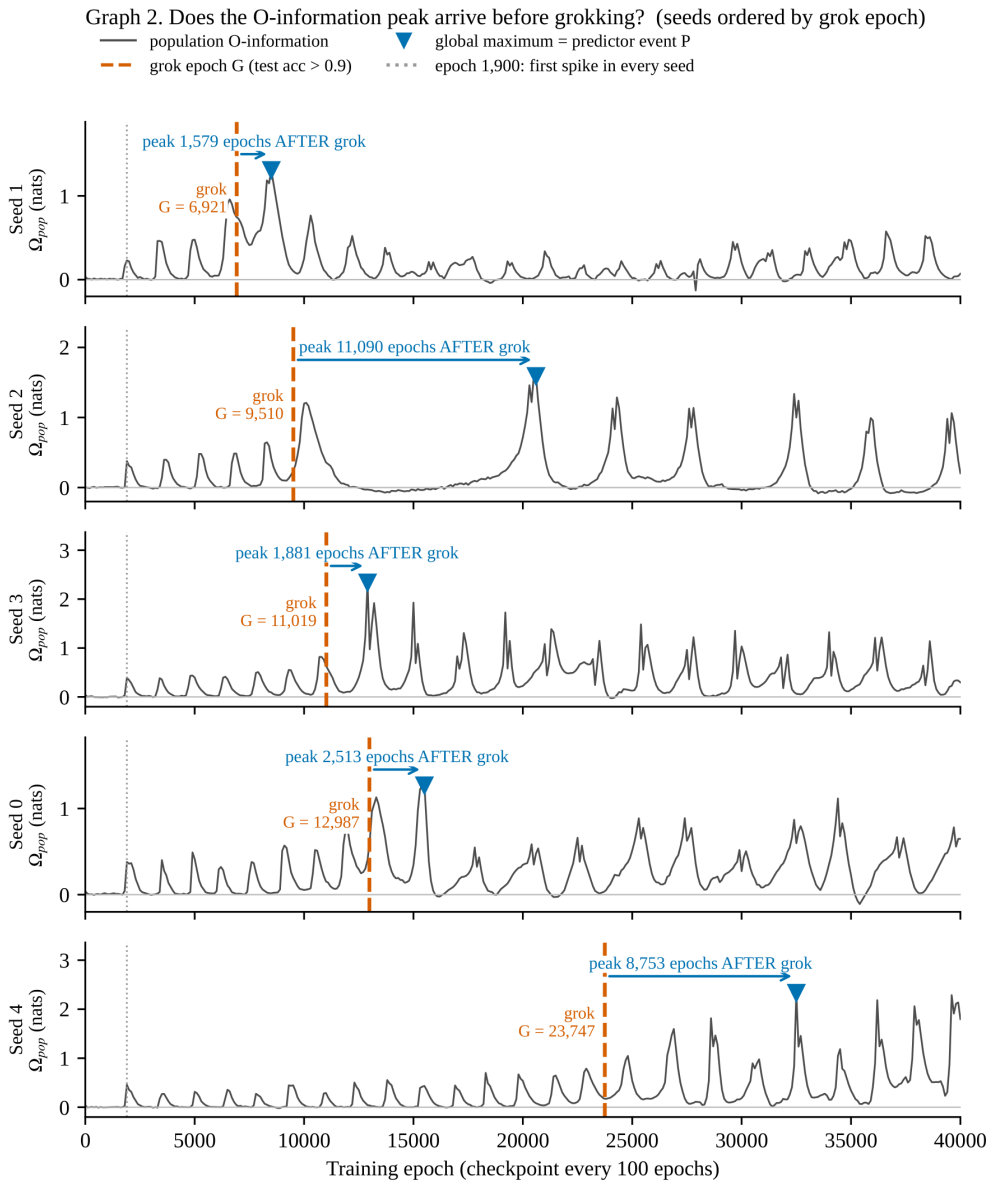
$$P = \arg\max_t \Omega_{\text{pop}}(t). \tag{10}$$

This event definition was fixed on 3 October 2026, before scoring. It is always defined and is the same type of landmark used for the other predictors in the benchmark. The grok epoch G is defined in Step A.

Observed population curves

The saved curves show the following results.

1. At epoch 0, Ω_{pop} ranges from approximately +0.002 to +0.040. Negative values are uncommon and small; the minimum observed value is −0.130.
2. The curves have repeated spikes and falls both before and after grokking, rather than one isolated transition.
3. In every seed, Ω_{pop} first exceeds 0.2 at epoch 1,900. Before grokking, subsequent spikes are spaced by roughly 1,400–1,600 epochs.
4. The global maximum occurs after the grok epoch in every seed.
5. The maximum values differ between seeds. Seed 4 rises from about 0.04 to 1.80, whereas seed 1 ends near its initial level (approximately 0.033 to 0.070).



Saved $\Omega_{\text{pop}}(t)$ for all five seeds, all 401 checkpoints, no smoothing. Each panel uses its own vertical range. Orange dashed line: grok epoch G. Blue triangle: global maximum P. Grey dotted line: epoch 1,900.

Figure 8: Original population O-information curves for all five seeds. Orange dashed lines indicate grok epochs, blue triangles indicate global maxima, and grey dotted lines mark epoch 1,900.

The shared onset at epoch 1,900 is earlier than the grok epoch in every run. This common timing suggests that the onset may be associated with general training dynamics, rather than specifically with grokking. This is an interpretation, not a tested causal explanation. The cause of the common onset was not investigated. The strongly negative initial values reported in the source paper are not reproduced in this implementation.

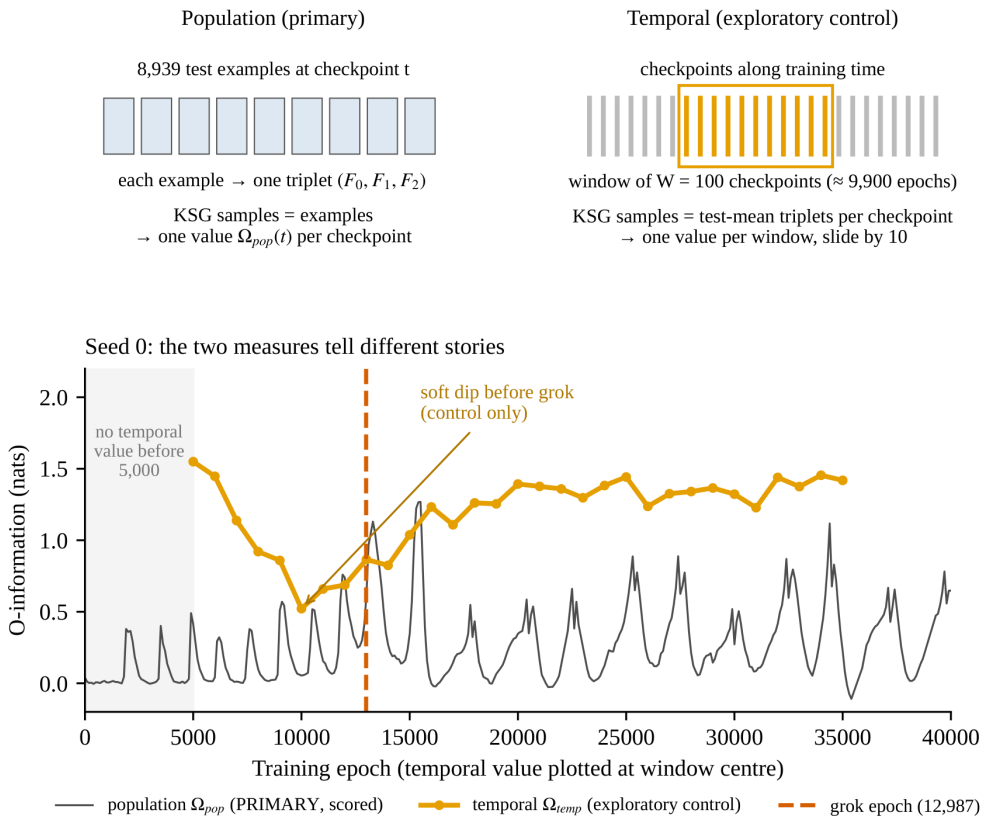
Population and temporal O-information

The population calculation has a potential confound: the three summaries are computed from the same forward pass, so common example difficulty may affect all of them. For example, an easy example may have a high F_0 and low F_2, while a difficult example may have F_0 close to F_1. This can create redundancy independently of grokking.

An exploratory temporal control was computed to examine a different sampling scheme. At each checkpoint, F_0, F_1, and F_2 are first averaged over the test set. The mean triplets are then collected in windows of 100 consecutive checkpoints, corresponding to approximately 9,900 epochs from first to last checkpoint in each

window. The window advances by 10 checkpoints. This gives 31 temporal values, plotted at centres from epoch 5,000 through epoch 35,000. No temporal value is available before epoch 5,000.

Graph 5. Population versus temporal O-information



Top: what one sample is in each measure. Bottom: both saved curves for seed 0. Each temporal value is plotted at its window centre.

Figure 9: Original comparison of population O-information and the exploratory temporal control for seed 0.

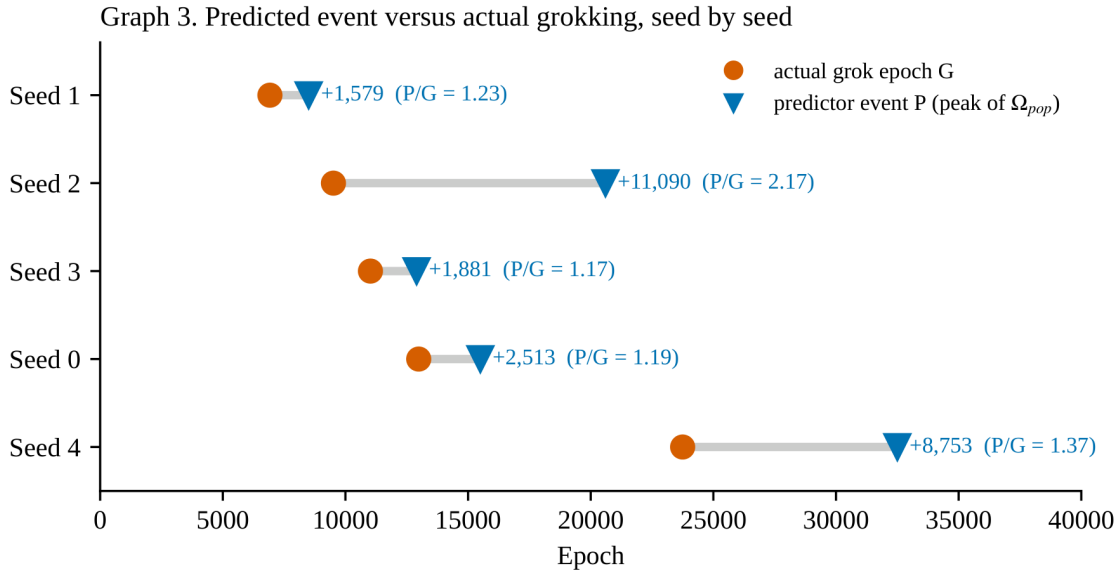
The temporal curve begins near 1.55, falls to about 0.52 near epoch 10,000, and then remains around 1.2–1.45. It is not negative in any seed. The population curve does not show the same dip. The different curve shapes indicate that population O-information is not simply identical to the temporal control. However, the control does not remove the difficulty confound, and the two calculations answer different questions. Because the temporal windows are wide and the control is not part of the frozen scoring protocol, the dip cannot be used as a precise early-warning event.

Peak timing by seed

Table 2 compares the grok epoch G with the peak epoch P. In this table, $P - G > 0$ means that the predictor event occurs after grokking.

Table 2: Grok and predictor event epochs.

Seed	G	P	$P - G$	P/G	$(G - P)/G$
0	12,987	15,500	2,513	1.19	−0.1935
1	6,921	8,500	1,579	1.23	−0.2281
2	9,510	20,600	11,090	2.17	−1.1661
3	11,019	12,900	1,881	1.17	−0.1707
4	23,747	32,500	8,753	1.37	−0.3686



Grok epoch G (orange circle) and peak epoch P (blue triangle) per seed. $P/G > 1$ means the peak came after grokking.

Figure 10: Original comparison of the grok epoch G and predictor event P for each seed.

Every peak occurs after grokking, between 1.17 and 2.17 times the corresponding grok epoch. Seed 1 has the earliest values of both G and P, and seed 4 has the latest. Seeds 2 and 3 change order between the two rankings. The peak timing therefore never provides an early warning, although the rank order has some agreement with the order of grok epochs.

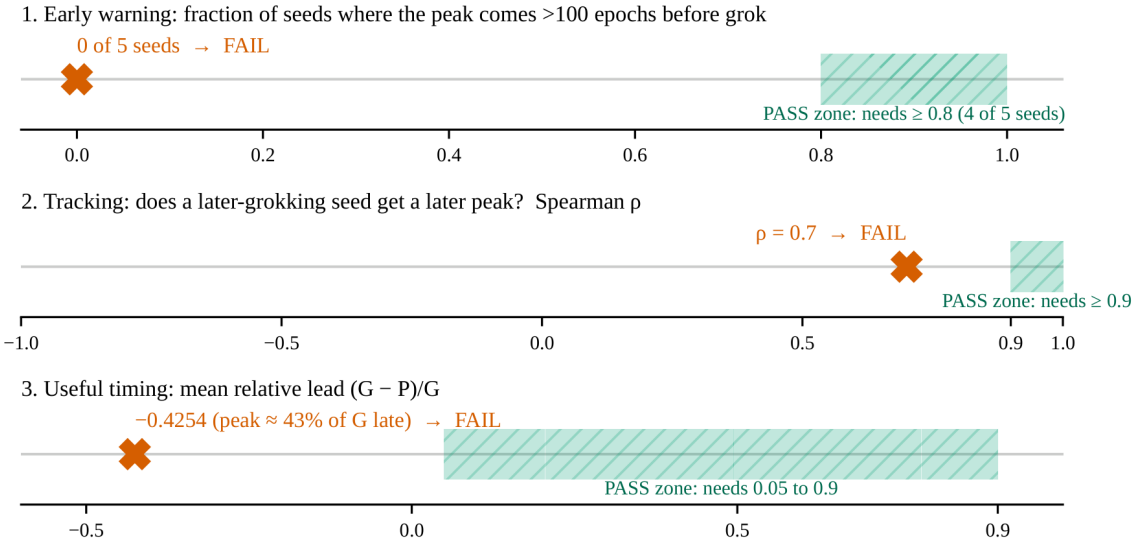
Frozen scoring protocol

All predictors in the benchmark are evaluated using the same three conditions. The protocol was fixed before scoring.

1. **Early-warning fraction:** a seed counts as a lead only if $P < G - 100$. The fraction must be at least 0.8, so at least four of the five seeds must lead.
2. **Rank tracking:** Spearman's rank correlation between the five G values and five P values must be at least 0.9. A value of 1 means that the rank orders agree exactly. The threshold was calibrated so random predictors exceed it about 1% of the time.
3. **Mean relative lead:** the mean of $(G - P)/G$ across seeds must be between 0.05 and 0.9. A negative value means the event is late. A value below 0.05 is too late; a value above 0.9 indicates an event very near the start of training, with little timing information.

The observed early-warning fraction is 0, the Spearman correlation is 0.7, and the mean relative lead is −0.4254. All three conditions fail.

Graph 6. The frozen-protocol scorecard: all three conditions must pass



Hatched green band: pass zone. Cross: Higher-MI's value.

Figure 11: Original scorecard showing the threshold for each of the three required conditions.

Step I. Conclusion and interpretation

The measured population O-information curve has repeated spikes, its first spike above 0.2 occurs at epoch 1,900 in every seed, and its global maximum occurs after grokking in all five seeds. Under the fixed protocol, Higher-MI is therefore classified as a failed early-warning predictor for this benchmark adaptation.

Comparison with the source paper

Table 3: Differences between the source-paper setting and this benchmark.

	Pomarico et al. (2025)	**This benchmark**
Model	Matrix product state classifier	One-layer transformer
Task	Three-class problems	Modular addition with 113 classes
Variables	Three class scores	Project-defined summaries (F_0, F_1, F_2)
Sample used for estimation	One training step; temporal sampling	One test example at a fixed checkpoint; population sampling
Claim tested here	Peak coincides with grokking in the paper's setting	Peak occurs before grokking

The source paper did not claim an early lead. The present benchmark uses a different model, task, set of variables, and sampling procedure. Its negative result does not refute the paper's result in its original setting.

Implementation corrections and checks

Two implementation errors were found and corrected before scoring.

1. **Strict neighbour counts.** The KSG calculation requires points strictly inside the distance threshold. The library call used an inclusive comparison, which included the neighbour defining the threshold. In one cross-check, about half the points were affected, and one O-information test value changed by a factor of about 2.6. Querying just below the threshold restored strict counting and matched a brute-force implementation bit for bit.

2. **Reproducible test split.** The recomputation code rebuilt the test set without resetting the seed. As a result, some pairs marked as test pairs were actually training pairs. The split is now rebuilt with the saved seed. The corrected seed-0 split contains 8,939 pairs and reproduces the stored population O-information values exactly.

The scoring protocol returns a pass for an ideal synthetic predictor in 100% of trials. Thus, the protocol can recognise a predictor designed to satisfy the criteria; Higher-MI fails under the observed data.

Assumptions, limitations, and open questions

- The reduction to (F_0, F_1, F_2) is an implementation choice and discards information from the original 113 logits.
- Samples for the primary measure are test examples at one checkpoint. The estimator uses $k = 4$; the source paper states this value for its transfer-entropy analysis.
- The "=" output is excluded, and the global-maximum rule is used to define the predictor event.
- Only five seeds are available. Spearman's ρ can take only 11 distinct values with five items. A random predictor constrained always to fire before grokking passes the rule about 21.7% of the time. This does not change the present result because the early-warning fraction is zero.
- Checkpoints are spaced 100 epochs apart, so faster changes cannot be observed.
- The temporal control is coarse: each window spans about 9,900 epochs, and no value is computed before epoch 5,000.
- Shared example difficulty may affect the population calculation. The temporal comparison partly addresses this concern but does not remove it.
- The common onset at epoch 1,900 remains unexplained.

Further work should use more seeds, save checkpoints more densely near epoch 1,900 and near each grok epoch, investigate the common onset, control for example difficulty, and test alternative reductions of the 113 logits.

Review questions and key definitions

1. **What is grokking?** Delayed generalisation. Training accuracy is near 0.99 by about epoch 205, while test accuracy exceeds 0.9 only between epochs 6,921 and 23,747 in these runs.
2. **What is a logit?** The raw, unnormalised score assigned to one of the 113 candidate answers. The largest logit determines the prediction.
3. **What are F_0, F_1, F_2?** The correct-answer logit, the largest incorrect-answer logit, and the mean logit of the remaining 111 incorrect answers.
4. **When is a prediction correct?** Exactly when $F_0 > F_1$.
5. **Why use three variables?** The O-information calculation is defined here for three variables. Reducing 113 logits to three is a project adaptation and loses information.
6. **Why use the test set?** Grokking is measured by generalisation to unseen examples; the training set is fitted much earlier.
7. **How is O-information interpreted?** Positive values indicate redundancy dominates under the stated sign convention; negative values indicate synergy dominates.
8. **What is $\Omega_{\text{pop}}(t)$?** The O-information estimate across test examples at a single fixed checkpoint.
9. **How does the temporal control differ?** It uses test-set mean triplets from checkpoints within sliding time windows, rather than individual test examples at one checkpoint.
10. **Why use the global maximum?** It is the fixed event definition adopted for this predictor in the benchmark.
11. **Why does this adaptation fail?** Its maximum occurs after grokking in all five seeds; the early-warning fraction, rank correlation, and mean relative lead all miss their thresholds.
12. **Does this show that the source paper is incorrect?** No. The model, task, variables, and sampling procedure differ, and the paper did not claim the early lead tested here.

Appendix A. Technical details

KSG estimator formulas

Distances are calculated using the maximum norm. For sample point i, let ε_i be its distance to the k-th nearest other point in the joint space. Let $N_x(i)$ denote the number of other points strictly within that distance in the X coordinate, with analogous definitions for N_y, N_z, N_{xz}, and N_{yz}. Thus, for a scalar X,

$$N_x(i) = \#\{j \neq i : |x_j - x_i| < \varepsilon_i\}. \tag{11}$$

With T samples, digamma function ψ, and $\langle\cdot\rangle$ denoting the average over sample points, the estimators used are

$$I(X;Y) = \psi(k) + \psi(T) - \left\langle \psi(N_x + 1) + \psi(N_y + 1) \right\rangle, \tag{12}$$

$$I(X;Y \mid Z) = \psi(k) + \left\langle \psi(N_z + 1) - \psi(N_{xz} + 1) - \psi(N_{yz} + 1) \right\rangle. \tag{13}$$

The population O-information estimate is the mean over all six permutations π of

$$I(F_{\pi_1}; F_{\pi_2}) - I(F_{\pi_1}; F_{\pi_2} \mid F_{\pi_3}). \tag{14}$$

At seed 0 and epoch 12,900, the six-ordering mean is 0.4044 nats, with standard deviation 0.0099 nats. The implementation also calculates the background mean as

$$F_2 = \frac{\sum_{j=0}^{112} \ell[j] - F_0 - F_1}{111}. \tag{15}$$

Scoring protocol and stored values

For the five seeds, the scoring metrics are

$$\text{lead fraction} = \frac{1}{5} \sum_{s=0}^{4} \mathbf{1}[P_s < G_s - 100], \tag{16}$$

$$\text{mean relative lead} = \frac{1}{5} \sum_{s=0}^{4} \frac{G_s - P_s}{G_s}, \tag{17}$$

$$\rho = \text{Spearman}(G, P). \tag{18}$$

The stored event values are shown in Table 2. The O-information values at the peak epochs are:

Seed	0	1	2	3	4
$\Omega_{\text{pop}}(P_s)$	1.2689	1.3089	1.5917	2.3403	2.3347

The ranks of G are $(4, 1, 2, 3, 5)$ and the ranks of P are $(3, 1, 4, 2, 5)$. The sum of squared rank differences is $\sum d_s^2 = 6$, giving

$$\rho = 1 - \frac{6 \sum_s d_s^2}{5(5^2 - 1)} = 1 - \frac{36}{120} = 0.7.$$

The Pearson correlation is 0.8976, but the protocol uses Spearman rank correlation. The threshold $\rho = 0.9$ is the 99th percentile over 10,000 random predictors with $P \sim U[0, 40000]$.

The protocol's synthetic-control pass rates are: ideal $P = 0.5G$, 100%; constant $P = 5{,}000$, 0%; $P = 1.5G$, 0%; noisy ideal, 91.04%; unrestricted random, 0.22%; and random predictors constrained to occur before grokking, 21.70%.

Exploratory signals (not included in the score)

The population zero-crossing events have $P_s = (500, 1600, 300, 1100, 200)$, lead fraction 1.0, $\rho = -0.7$, and mean relative lead 0.9181. This is an early signal near initialisation, similar to the failure mode of a constant-early predictor.

For the temporal-control maximum, $P_s = (5000, 5000, 19000, 5000, 14000)$, the lead fraction is 0.8, and the mean relative lead is 0.1703. Three of the apparent leads occur at the first available window centre. The stored correlation is 0.00 under position-based tie-breaking and 0.11 under tie-aware ranking; both are far below 0.9.

Transfer entropy was also explored for the test-set mean correct-answer and runner-up logits:

$$\mathrm{TE}_{X\to Y}(\tau) = I(Y(t+\tau); X(t) \mid Y(t)), \qquad \tau = 1, \ldots, 10. \tag{19}$$

At $\tau = 1$, the estimate from the correct-answer logit to the runner-up logit is larger than the reverse direction in every seed. The trajectories have strong trends, so this result is not evidence of a causal relationship.

Model and training configuration

The model and training settings are:

- one-layer transformer;
- four attention heads, each of width 32;
- model dimension $d_{\text{model}} = 128$;
- MLP width 512 with ReLU activation;
- no LayerNorm; learned positional embeddings; untied embeddings;
- AdamW optimiser, learning rate 10^{-3}, weight decay 1.0, $\beta = (0.9, 0.98)$;
- full-batch training for 40,000 epochs.

The shared model file and training loop were not modified for the Higher-MI experiment.

Source files and reproduction

The stored curves and scores are located in

`08_Experiments/results/nanda_unified/`

Relevant files include `seed_<N>/higher_mi/*`, `aggregate_higher_mi.json`, and `seed_<N>/training/*_acc_history.npy`.

The recomputation commands used were:

```
python 06_Code/scripts/run_nanda_benchmark.py \\
    --predictors higher_mi --seeds 5 --overwrite higher_mi \\
    --output_dir 08_Experiments/results/nanda_unified
python 06_Code/scripts/analyze_nanda_unified.py
```

A full recomputation took approximately 27 minutes in the recorded run. The thesis text rounds the mean relative lead to −0.43 in some places; the stored value used for scoring is −0.4254.

Reference

Pomarico et al. (2025). Source paper identified in the project as arXiv:2507.23346.